

Muhammad Adnan Shahzad

Machine Learning & AI Researcher

Montreal, QC, Canada

+1 438 466 3377

✉ muhammad.shahzad@unicam.it

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Profile

Interdisciplinary researcher with a **PhD in Physics** and current training in Computer Science, focusing on **Machine Learning, Generative Models, and Data-Driven Systems**. Strong background in stochastic processes, numerical simulation, and optimization, with hands-on experience in deep learning, computer vision, and large-scale ML pipelines. Recently completed three end-to-end production-grade ML projects spanning traditional ML, computer vision, and RAG systems.

Education

- 2023–2026 **B.Sc. in Computer Science (In Progress)**, *Concordia University*, Montreal, QC
Expected Fall 2026
Relevant Coursework: Machine Learning, Pattern Recognition, Computer Vision, Artificial Intelligence, Big Data Analytics
- 2012–2015 **PhD in Physics**, *University of Camerino*, Italy
Thesis: *Protein Translocation in Narrow Pores*
Focus: Molecular Dynamics, Stochastic Processes, Statistical Mechanics

Research Interests

Machine Learning, Deep Learning, Generative Models, Optimization, Representation Learning, Scientific Machine Learning, Computer Vision, Retrieval-Augmented Generation

Selected Projects

2026 Deep Learning Detection of Beyond-GR Deviations in Gravitational Waves

- Built a complete machine-learning pipeline to detect controlled deviations from General Relativity in gravitational-wave signals, using synthetic aLIGO noise and real LIGO H1 detector strain from GWOSC
- Designed a hybrid classifier combining a 1D convolutional neural network on raw 4-second waveforms (16,384 samples) with 10 hand-crafted statistics (mean $|h|$, std, RMS, max, p95, p99, p99.9, p99.99, skewness, kurtosis); total 31,697 parameters
- Tested three deviation families (amplitude, phase, frequency modulation) parameterized by a dimensionless strength coefficient β
- Measured a quantitative detectability curve using the real GW150914 strain as a template, finding a detection threshold at $\beta \approx 0.25$ with accuracy rising smoothly from chance ($\beta \leq 0.2$) to perfect classification ($\beta \geq 0.5$)
- Demonstrated generalization to unseen deviation types (88–100% accuracy) and to binary mass ratios $q = 1$ to $q = 10$ (89–93% accuracy)
- Quantified the limits of ML-only beyond-GR searches: at physically realistic deviation strengths ($\beta \sim 10^{-3}$ – 10^{-2}), the classifier fails, motivating hybrid ML + matched-filter approaches
- Authored full LaTeX manuscript (submitted for publication, 2026); all code and reproducible pipeline available on GitHub
- **Technical Stack:** PyCBC, LALSuite, GWpy, PyTorch, NumPy, SciPy, scikit-learn, Matplotlib, LaTeX
- **Key Techniques:** 1D Convolutional Neural Networks, Hybrid Feature Engineering, Whitening with Welch PSD Estimation, Butterworth Bandpass Filtering, Amplitude-Controlled Classification, ROC-AUC, Feature-Importance Analysis, Random Forest Baseline

2026 Kaggle & AI Competitions — 7 Solutions (Top 15–20%)

- Completed 7 Kaggle/AI competitions across classification, regression, deep learning, and autonomous agents with ensemble methods
- **Electric Vehicle Purchase Prediction:** Achieved 0.94170 AUC (Top 15%) using 5-Fold CV + OOF + Rank Averaging ensemble of XGBoost, LightGBM, and CatBoost
- **Customer Churn Prediction:** Achieved 0.91258 AUC (Top 15–20%) with ensemble of XGBoost, LightGBM, CatBoost, Random Forest, and Logistic Regression
- **ARC-AGI-3 (ARC Prize 2026):** Built exact model-based BFS planner for LS20 game; solved Level 1 in 13 actions (beating human baseline of 22); Score: 3.57
- **Spaceship Titanic:** Achieved 0.79962 Accuracy (Top 15–20%) with tuned LightGBM and engineered features (cabin deck, group size, total spend, CryoSleep)
- **Titanic:** Achieved 0.8268 Validation Accuracy (0.76794 Public LB, Top 20%) using PyTorch Deep Learning with custom neural network architecture
- **House Prices:** Achieved 0.12669 RMSE (Top 15–20%) using CatBoost with Optuna hyperparameter tuning and advanced feature engineering
- **Autonomous Agent Prediction:** Built LLM-powered agent framework for autonomous ML workflows achieving 0.794 AUC
- **Technical Stack:** Python, PyTorch, XGBoost, LightGBM, CatBoost, Optuna, Scikit-learn, Pandas, NumPy, BFS Planning, State Space Search
- **Key Techniques:** 5-Fold Cross-Validation, OOF Predictions, Rank Averaging, Feature Engineering, Hyperparameter Tuning, Ensemble Methods, Deep Learning, Model-Based BFS Planning, State Fingerprinting, Transition Learning

2026 **LLM Serving Platform with Kubernetes & Observability**

- Built production-ready LLM inference platform with FastAPI, Ollama, Redis caching, and Kubernetes orchestration
- Implemented 2+ replicas with horizontal pod autoscaling (HPA) achieving 100% cache hit rate under load
- Integrated Prometheus metrics (latency, tokens, requests) and Grafana dashboards for real-time monitoring
- Containerized with Docker and deployed on Kind cluster with service discovery and load balancing
- Load-tested with 10+ concurrent users: 21.8ms avg latency, 100% cache hit rate, 600+ tokens throughput

2026 **ML Monitoring & Observability on Kubernetes**

- Deployed complete ML monitoring system with drift detection on Kubernetes (kind cluster)
- Implemented FastAPI with 2 replicas, service discovery, and load balancing
- Integrated MLflow for experiment tracking and model versioning
- Added Prometheus for metrics collection and Grafana for visualization
- Built statistical drift detection monitoring predictions in real-time
- Demonstrates production-grade MLOps with Kubernetes, observability, and monitoring

2026 **Semantic Search System with Hybrid Retrieval**

- Built comprehensive semantic search system comparing keyword (BM25), vector (FAISS), hybrid, and reranked approaches
- Implemented sentence embeddings with all-MiniLM-L6-v2 for document representation
- Achieved 20% improvement in NDCG@5 using hybrid search over pure keyword or vector search
- Added cross-encoder reranking with ms-marco-MiniLM-L-6-v2 for further quality improvement
- Evaluated using standard IR metrics: Recall@K, Precision@K, MRR, and NDCG
- Demonstrates search engineering, IR evaluation, and embedding-based retrieval

2026 **Advanced Agentic RAG System with Multi-Tool Support**

- Built multi-tool agentic RAG system with LangGraph, RAG, SQL, web search, and calculator tools
- Implemented agent memory and state management for complex multi-step reasoning
- Created vector store with Chroma for document retrieval and similarity search
- Integrated comprehensive evaluation metrics: retrieval accuracy, answer correctness, and hallucination rate
- Deployed with FastAPI and Docker support

2026 **Kubernetes Deployment with kind and FastAPI**

- Deployed FastAPI application on local Kubernetes cluster using kind (Kubernetes in Docker)
- Implemented complete Kubernetes manifests with Deployments, Services, ConfigMaps, Secrets, and HPA
- Configured horizontal pod autoscaling and rolling update strategies
- Managed 3-node cluster with kubectl for orchestration and monitoring
- Demonstrates container orchestration, cloud-native patterns, and Kubernetes operations

2026 **Data Engineering + ML Pipeline (Live API)**

- Built end-to-end data pipeline with Kafka, Spark, Airflow, PostgreSQL, and MLflow
- Deployed REST API on Render.com with real-time prediction endpoints
- Implemented streaming data processing with PySpark and orchestration with Airflow
- Integrated MLflow for experiment tracking and model versioning
- Demonstrates full-stack ML engineering: ingestion, processing, orchestration, and serving
- Live API: <https://data-engineering-pipeline.onrender.com>

- 2026 **Production Recommendation System with Hybrid Filtering**
- Built a production-ready recommendation system combining collaborative and content-based filtering using SVD++ (RMSE: 0.9314) on the MovieLens dataset
 - Developed a hybrid recommendation algorithm that integrates collaborative filtering with TF-IDF genre similarity for improved personalization
 - Created a RESTful API with FastAPI featuring interactive Swagger documentation, user-based recommendations, similar-item search, and rating predictions
 - Containerized the application with Docker and deployed on Render.com with a live public API endpoint
 - Designed a modular, extensible architecture supporting collaborative, content-based, and hybrid recommendation models for easy future enhancements
- 2026 **End-to-End ML Pipeline & API Deployment (House Price Prediction)**
- Built production-ready ML pipeline with data cleaning, feature engineering, model training, and hyperparameter tuning using scikit-learn and XGBoost
 - Achieved **R² score of 0.8354** with tuned Gradient Boosting model
 - Developed REST API using FastAPI with interactive Swagger documentation
 - Containerized with Docker and deployed on Render.com (live demo available)
 - Demonstrates full MLOps lifecycle: model serialization, version control, and cloud deployment
- 2026 **Real-Time Object Detection with YOLOv11 & Gradio**
- Implemented YOLOv11 object detection for safety equipment monitoring (helmets, vests, people)
 - Built interactive web interface with Gradio supporting image and video processing
 - Achieved real-time detection with confidence scoring and color-coded bounding boxes
 - Deployed on Hugging Face Spaces with permanent live URL
 - Demonstrated computer vision, model inference optimization, and web deployment
- 2026 **RAG Chatbot with TinyLlama & PDF Support**
- Developed Retrieval-Augmented Generation (RAG) chatbot using TinyLlama (637MB LLM)
 - Implemented document chunking, keyword-based retrieval, and LLM generation
 - Added support for PDF documents (tested with Bishop's Pattern Recognition and ML book)
 - Built Gradio 6.x web interface with chat history and example questions
 - Deployed on Hugging Face Spaces with live demo
 - Showcases LLM integration, RAG pipeline design, and production deployment
- 2026 **Production MLOps Pipeline with CI/CD and MLflow**
- Built end-to-end MLOps pipeline with experiment tracking (MLflow), automated testing (pytest), and CI/CD (GitHub Actions)
 - Implemented automated model training, testing, and deployment with GitHub Actions
 - Created comprehensive unit tests (4/4 passing) for model validation and quality assurance
 - Integrated MLflow for experiment tracking with SQLite backend (3 models tracked: Linear-Regression, RandomForest, GradientBoosting)
 - Achieved best model performance: RandomForest with $R^2=0.8051$
 - Automated workflow triggers on every push, ensuring code quality and reproducibility
 - Containerized with Docker and Docker Compose for multi-service deployment
 - Demonstrates professional MLOps: reproducibility, testing, monitoring, and automation
- 2026 **Visual Question Answering with BLIP-VQA**
- Built VQA system using BLIP-VQA-Base for image understanding
 - Integrated Hugging Face API for zero-shot inference
 - Developed Gradio web interface with image upload and question input
 - Deployed on Hugging Face Spaces with permanent URL

- 2025 **Hybrid Quantum-Classical Neural Networks for Image Classification**
- Proposed and implemented hybrid quantum-classical architectures combining parameterized quantum circuits with deep CNNs (PyTorch, PennyLane)
 - Benchmarked against classical models on MNIST, CIFAR100, and STL10, achieving consistent accuracy improvements (+9–10% on complex datasets)
 - Demonstrated **5–12× faster training** and reduced model size (up to 32% fewer parameters)
 - Performed robustness evaluation under adversarial perturbations ($\epsilon = 0.1$), showing enhanced stability
 - Documented results in an arXiv submission with reproducible code, datasets, and LaTeX report
- 2025 **Multi-Source Financial Sentiment Analysis**
- Built end-to-end ML pipeline integrating stock, news, and social media data
 - Reduced 1795 features to 35 using PCA (95% variance retained)
 - Achieved up to **70.3% accuracy** using ensemble learning
- 2026 **Plant Disease Classification (CNNs)**
- Benchmarked multiple CNN architectures on 39-class dataset
 - Achieved **97.15% accuracy** using ResNet18
 - Analyzed limitations of transfer learning in domain-specific datasets
- 2025 **FFHQ Latent Diffusion Model – Analysis of Pretrained Generative Model**
- Analyzed a pretrained Latent Diffusion Model (FFHQ) with **99.8% parameter consistency** after resolving configuration mismatches
 - Performed GPU benchmarking and identified sub-optimal scaling behavior (0.55 img/s, batch size = 4)
 - Developed a multi-method diversity evaluation framework (PCA, t-SNE, UMAP, manifold learning)
 - Discovered critical mode collapse (avg similarity ≈ 0.988 , novelty = 0%) via multi-seed experiments
 - Investigated latent space structure, showing extremely low intrinsic dimensionality and limited distribution coverage
- 2026 **AI Agent with Tool Integration**
- Built an autonomous AI agent with tool-calling capabilities using TinyLlama and LangGraph
 - Integrated functional tools including a calculator, time/date, and weather API
 - Developed an interactive Gradio web interface for seamless user interaction
 - Deployed the application as a live, permanent demo on Hugging Face Spaces
 - Implemented dynamic tool selection and prompt engineering for accurate agent reasoning
 - Demonstrates practical expertise in LLM agents, tool design, and full-stack deployment
- 2026 **Rust Backend API with SQLx & Actix-Web**
- Built production REST API using Rust, Actix-Web, and SQLx with compile-time SQL verification
 - Designed PostgreSQL database with repository pattern and type-safe database operations
 - Implemented 20+ API endpoints with automatic query validation via `cargo sqlx prepare`
 - Integrated analytics features including download tracking, member statistics, and real-time reporting
 - Demonstrates systems programming, database architecture, and production-grade Rust development
- 2026 **Emacs-Based ML/Data Science Workflow System**
- Developed integrated Python/R workflow using Emacs, Org-mode, and Jupyter
 - Enabled reproducible research combining code, documentation, and visualization
 - Implemented intelligent auto-completion (LSP, Pyright, Anaconda) and inline plotting
 - Automated Jupyter-to-Org conversion for efficient academic and ML workflows

Publications

Shahzad, M.A., *Hybrid Quantum-Classical Model for Image Classification*, arXiv (2025)
Cecconi, F., Shahzad, M.A., et al., *Frequency-Control of Protein Translocation Across an Oscillating Nanopore*, Physical Chemistry Chemical Physics (RSC), 2017

Technical Skills

ML/DL Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, CatBoost, Hugging Face Transformers, LangChain, LangGraph
Computer Vision	OpenCV, YOLOv11, Ultralytics, CNNs (ResNet, EfficientNet, VGG), Image Classification, Object Detection, Video Processing, Transfer Learning, Vision Transformers, BLIP-2, Multimodal AI
Natural Language Processing & LLMs	Transformers, Tokenization, Text Embeddings, RAG (Retrieval-Augmented Generation), Document Processing, PDF Extraction, NLTK, SpaCy, Ollama (TinyLlama, Llama3.2), Agentic RAG, Tool Calling, Prompt Engineering
Generative AI	Latent Diffusion Models, Stable Diffusion, VAE, GANs, Diffusion Models, Image Generation, Mode Collapse Analysis
Quantum ML	PennyLane, Quantum Circuits, Variational Quantum Algorithms, Hybrid Quantum-Classical Models
Search & Information Retrieval	BM25, FAISS, Sentence Transformers, Cross-encoders, Hybrid Search, Semantic Search, Vector Databases (Chroma, FAISS), Similarity Search, Reranking
Data Science & Analytics	Pandas, NumPy, SciPy, Data Cleaning, Feature Engineering, Exploratory Data Analysis (EDA), Statistical Analysis, Dimensionality Reduction (PCA, t-SNE, UMAP)
MLOps & Deployment	FastAPI, Docker, Gradio, Hugging Face Spaces, Render.com, REST APIs, Swagger/OpenAPI, Model Serialization (Pickle, Joblib), Containerization, Kubernetes, kind, Helm, CI/CD (GitHub Actions), MLflow, Evidently, Prometheus, Grafana
Data Engineering	Apache Kafka, Apache Spark (PySpark), Apache Airflow, PostgreSQL, SQLAlchemy, ETL Pipelines, Stream Processing, Real-time Data Pipelines, DVC (Data Version Control)
Monitoring & Observability	Prometheus, Grafana, Evidently, Drift Detection, Model Performance Monitoring, Data Quality Monitoring, Statistical Drift Detection
Programming Languages	Python (Advanced), C++, Java, SQL, Bash/Shell Scripting
Version Control & CI/CD	Git, GitHub, Git LFS, GitHub Actions, CI/CD Pipelines, Automated Testing (pytest), Docker Compose
Scientific Computing	MATLAB, Molecular Dynamics (GROMACS, LAMMPS), Numerical Simulation, Stochastic Processes, Optimization Algorithms
Development Tools	Jupyter Notebook, Emacs, VS Code, LaTeX, Org-mode, Linux/Unix, Conda, Poetry, Makefile
Database & Big Data	SQL (PostgreSQL, SQLite), NoSQL, Data Warehousing, ETL Pipelines, Chroma (Vector DB)

Cloud & Infrastructure AWS, Google Cloud Platform, Render, Hugging Face Infrastructure, Docker Hub

Agentic AI LangGraph, Tool Calling, Agent Memory/State, Multi-step Reasoning, RAG Agents, SQL Agents, Web Search Agents, Agent Evaluation

Research Experience

2012–2015 **PhD Researcher**, *University of Camerino*

- Conducted molecular dynamics simulations of protein translocation
- Studied stochastic transport processes and frequency-driven dynamics
- Analyzed large-scale simulation data and developed computational models

Experience

2017–2019 **Assistant Professor**, *Pakistan*

- Taught physics and computational methods
- Supervised student projects in simulation and data analysis

Additional

Tools LaTeX, Jupyter, Emacs, Docker, Git, VS Code, Linux, Bash, Make, kubectl, kind, Helm

Interests Machine Learning Research, Scientific Computing, MLOps, LLMs, Quantum Computing, AI Agents, Information Retrieval

Languages English (Fluent), Urdu (Native), Italian (Basic)